

Bayesian Credit Risk

Applied Statistical Modelling

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Credit Risk Data

The Dataset

- ▶ *Credit Risk Dataset*, published on Kaggle.
- ▶ Originally $n = 32,581$ loan applications; target $Y_i \in \{0, 1\}$ (`loan_status`: 1 = default).
- ▶ After cleaning (implausible age/emp. length, missing values): $n = 28,632$.
- ▶ Predictors excluded: `loan_grade`, `loan_int_rate` (post-hoc lender variables), `cred_hist_length` ($r \approx 0.88$ with age).
- ▶ `person_income` log-transformed, continuous vars standardized, categorical dummy-coded.
- ▶ Final design matrix: $n = 28,632$, $p = 13$ candidate predictors.

Logistic Regression

- ▶ Binary response: probability of loan default.

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad i = 1, \dots, n$$

$$\text{logit}(\pi_i) = \eta_i = \beta_0 + \boldsymbol{\beta}^T \mathbf{x}_i$$

- ▶ Parameter vector $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$ plus intercept β_0 .
- ▶ Not conjugate $\Rightarrow$ MCMC to sample from the posterior.

Spike & Slab Prior

- ▶ Spike & Slab prior for **posterior variable selection**:

$$\beta_j \mid \gamma_j \sim (1 - \gamma_j) N(0, \tau^2) + \gamma_j N(0, cc \cdot \tau^2)$$

$$\gamma_j \mid \theta_j \sim \text{Bernoulli}(\theta_j) \quad \theta_j \sim \text{Unif}(0, 1)$$

- ▶ Narrow **spike** (variance τ^2) vs wide **slab** (variance $cc \cdot \tau^2$); intercept $\beta_0 \sim N(0, 10^2)$, not selected.
- ▶ Settings: $c = 100$, intersection $k = 0.05 \Rightarrow \tau^2 \approx 2.71 \times 10^{-4}$, $cc \cdot \tau^2 \approx 2.71$.
- ▶ Each θ_j estimated per variable: the data set the inclusion odds of every predictor.

Model Specification in NIMBLE

```
code_logit <-nimbleCode({  
  for (i in 1:n) {  
    Y[i] ~dbern(eta[i])  
    logit(eta[i]) <-beta0 + inprod(X[i, 1:p], beta[1:p])  
  }  
  beta0 ~dnorm(0, sd = 10)  
  for (j in 1:p){  
    sig2[j] <-equals(gamma[j], 0) * var_spike +  
              equals(gamma[j], 1) * var_slab  
    beta[j] ~dnorm(0, sd = sqrt(sig2[j]))  
    gamma[j] ~dbern(theta[j])  
    theta[j] ~dunif(0, 1)  
  }  
  var_spike <-tau2  
  var_slab <-cc * tau2  
})
```

- Iterations = 80,000, burn-in = 10,000, thinning = 20, chains = 3
⇒ 10,500 samples.

MCMC Summary Table I

Coefficients

Variable	Mean	SD	2.5%	Median	97.5%	$\hat{R}$	n.eff
Intercept (β_0)	-1.586	0.047	-1.681	-1.585	-1.494	1.00	3,366
person_age	-0.008	0.012	-0.032	-0.008	0.016	1.00	10,723
person_emp.length	-0.015	0.013	-0.041	-0.015	0.010	1.00	10,241
loan_amnt	0.306	0.043	0.220	0.307	0.391	1.00	2,908
loan_percent_income	0.594	0.038	0.519	0.594	0.669	1.00	2,994
log_income	-0.716	0.041	-0.799	-0.716	-0.636	1.00	3,056
home_ownershipMORTGAGE	-0.793	0.039	-0.870	-0.792	-0.716	1.00	10,133
home_ownershipOWN	-2.099	0.090	-2.275	-2.099	-1.925	1.00	10,500
loan_intentDEBTCONSOLIDATION	0.526	0.058	0.411	0.526	0.642	1.00	4,549
loan_intentEDUCATION	-0.240	0.061	-0.357	-0.242	-0.119	1.00	3,968
loan_intentHOMEIMPROVEMENT	0.672	0.064	0.546	0.671	0.800	1.00	4,933
loan_intentMEDICAL	0.358	0.057	0.247	0.358	0.470	1.00	4,343
loan_intentVENTURE	-0.410	0.063	-0.533	-0.411	-0.287	1.00	4,832
cb_person_default_on_fileY	1.128	0.040	1.049	1.128	1.206	1.00	10,137

MCMC Summary Table II

Coefficients

Inclusion indicators (γ , i.e. PIP)

Variable	PIP	SD	$\hat{R}$	n.eff
person_age	0.016	0.124	1.01	10,086
person_emp.length	0.034	0.180	1.01	9,692
loan_amnt	1.000	0.000	NaN	0
loan_percent_income	1.000	0.000	NaN	0
log_income	1.000	0.000	NaN	0
home_ownershipMORTGAGE	1.000	0.000	NaN	0
home_ownershipOWN	1.000	0.000	NaN	0
loan_intentDEBTCONSOLIDATION	1.000	0.000	NaN	0
loan_intentEDUCATION	0.992	0.089	1.03	1,580
loan_intentHOMEIMPROVEMENT	1.000	0.000	NaN	0
loan_intentMEDICAL	1.000	0.000	NaN	0
loan_intentVENTURE	1.000	0.000	NaN	0
cb_person_default_on_fileY	1.000	0.000	NaN	0

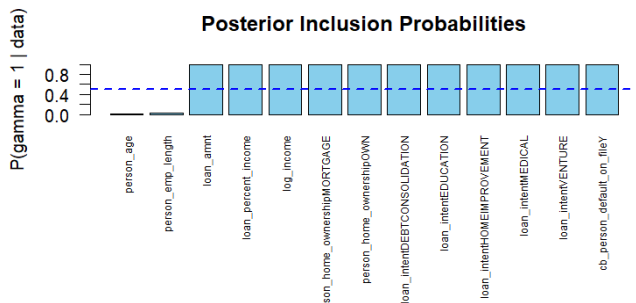
MCMC Convergence

- ▶ All $\hat{R} = 1.00$ for the β 's; ESS from $\sim 2,900$ to $> 10,700$: good convergence, low autocorrelation.
- ▶ For the γ_j : $\hat{R}$ defined only for the 3 variables that occasionally switch (person_age, person_emp_length, loan_intentEDUCATION); NaN for the other 10.
- ▶ Geweke: only 2 of 81 cells with $|z| > 1.96$, fewer than the 4 expected by chance.
- ▶ Traceplots stationary, ACF decaying rapidly.

▶ Traceplots & ACF

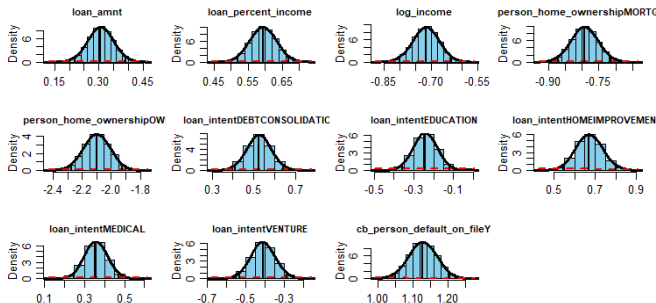
Variable Selection – PIP & MPM

- ▶ $\text{PIP} = \mathbb{E}[\gamma_j \mid \text{data}]$; Median Probability Model selects $\text{PIP} > 0.5$.



- ▶ Excluded: `person_age` (0.016), `person_emp_length` (0.034).

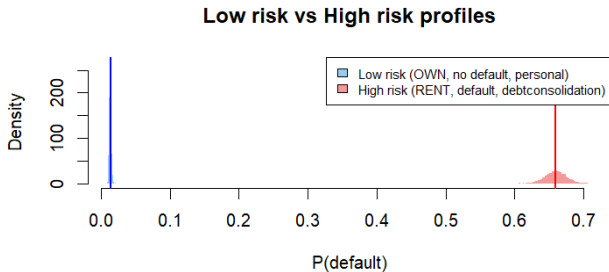
Posterior Distributions of Selected Effects



- ▶ Posteriors approximately Normal, credible intervals excluding zero; slab prior flat over each posterior (data-dominated).
- ▶ Strongest protective effect: `home_ownershipOWN` (≈ -2.10); strongest risk: `cb_default_on_fileY` ($\approx +1.13$).

Low Risk vs High Risk

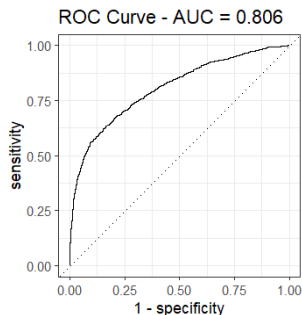
- ▶ **Low risk profile:** home owner (OWN), no prior default, personal loan (reference category), below-average loan-to-income ratio (-1 SD); age, income, loan amount, and employment length at the sample mean.
- ▶ **High risk profile:** renter (RENT), prior default on file, debt-consolidation loan, above-average loan-to-income ratio ($+1$ SD); same average values for the remaining continuous predictors.



▶ Posterior mean $P(\text{default})$: **0.0137** (low risk) vs **0.6505** (high risk).

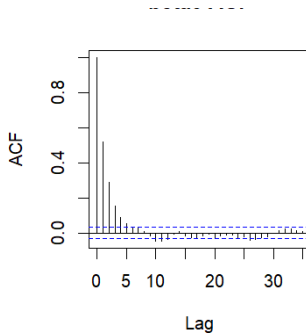
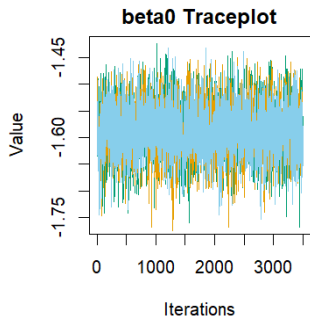
Predictive Validation

- ▶ Frequentist logistic regression on the 11 MPM-selected predictors, 80/20 split.
- ▶ Test-set **AUC** = 0.806: strong out of sample discrimination.



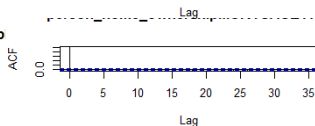
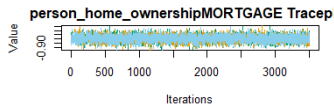
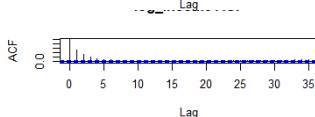
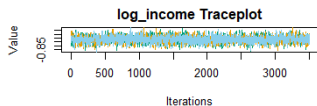
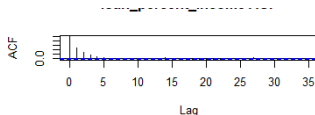
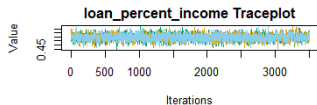
**THANKS FOR YOUR
ATTENTION!!!**

Traceplots and ACF



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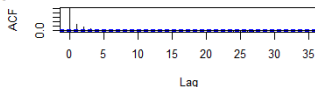
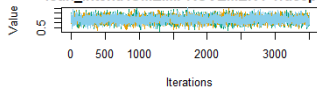
Traceplots and ACF



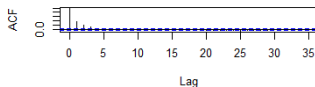
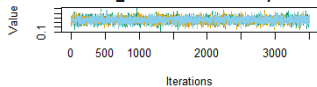
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Traceplots and ACF

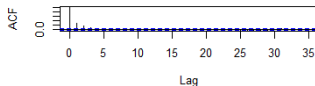
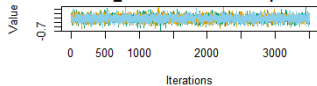
loan_intentHOMEIMPROVEMENT Traceplot



loan_intentMEDICAL Traceplot

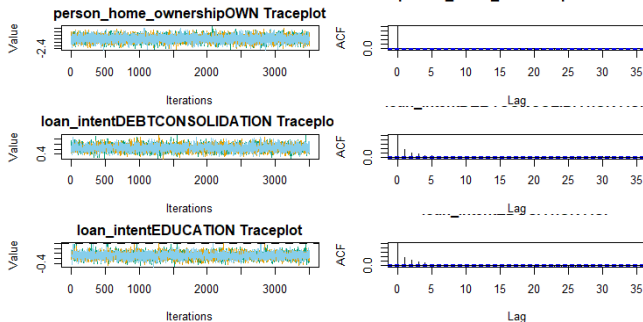


loan_intentVENTURE Traceplot



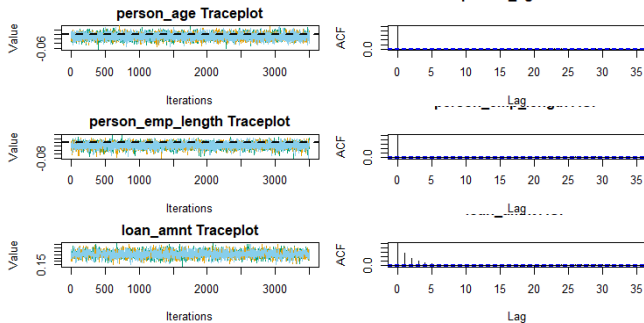
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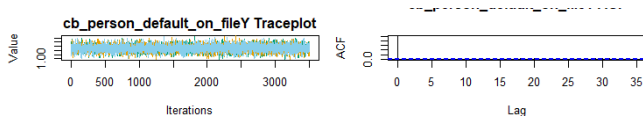
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